

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2026

Duration: 30 minutes

Quiz 3

Total: 20 marks

Name:	ID:	1 Points
Section:		1 Points

Pumping Lemma (8 Marks)

$$L = \{0^n 1^{2n} \mid n \geq 0\}$$

Prove that L is **not regular** using the Pumping Lemma.

Assume, for contradiction, that L is regular. Let p be the pumping length.

Choose the string

$$w = 0^{p^2} 1^{2p}.$$

Clearly, $w \in L$ by taking $n = p$, and $|w| \geq p$.

By the Pumping Lemma, we can write

$$w = xyz$$

such that

$$|xy| \leq p, \quad |y| > 0,$$

and

$$xy^i z \in L \quad \text{for every } i \geq 0.$$

Since the first p symbols of w are all 0's, y must contain only 0's. Thus,

$$y = 0^k$$

for some $1 \leq k \leq p$.

Now pump down by choosing $i = 0$. Then

$$xy^0 z = xz = 0^{p^2-k} 1^{2p}.$$

The string still contains $2p$ ones. Therefore, if $xz \in L$, we must have $n = p$. But a string corresponding to $n = p$ must contain exactly p^2 zeros, whereas xz contains only $p^2 - k$ zeros.

Hence,

$$xz \notin L.$$

This contradicts the Pumping Lemma. Therefore,

L is not regular.

Bonus (2 Marks, you may answer this after completing the rest)

Use the pumping lemma to prove that the following language is not regular:

$L(M) \rightarrow \{w_1 1 w_2 \mid w_1 \text{ is a substring of } w_2 \text{ and } |w_1| \neq |w_2|\}$, where $\Sigma = \{0, 1\}$.

Assume, for contradiction, that L is regular. Let p be the pumping length.

Choose

$$w = 0^p 1 0^{p+1}.$$

Here,

$$w_1 = 0^p \quad \text{and} \quad w_2 = 0^{p+1}.$$

Clearly, w_1 is a substring of w_2 , and

$$|w_1| = p \neq p + 1 = |w_2|.$$

Therefore, $w \in L$.

By the Pumping Lemma, w can be divided as

$$w = xyz$$

such that

$$|xy| \leq p, \quad |y| > 0,$$

and $xy^iz \in L$ for every $i \geq 0$.

Because the first p symbols are all 0's, y must have the form

$$y = 0^k$$

for some $1 \leq k \leq p$.

Now pump y by choosing $i = 2$:

$$xy^2z = 0^{p+k}10^{p+1}.$$

There are two cases:

1. If $k = 1$, then

$$|w_1| = p + 1 = |w_2|,$$

which violates the condition $|w_1| \neq |w_2|$.

2. If $k \geq 2$, then

$$|w_1| = p + k > p + 1 = |w_2|.$$

Therefore, w_1 cannot be a substring of the shorter string w_2 .

Thus, in every case,

$$xy^2z \notin L.$$

This contradicts the Pumping Lemma. Hence,

L is not regular.

CFG (10 Marks)

$L_1 \rightarrow \{w \in \{0, 1\}^* \mid w \text{ contains equal numbers of 0s and 1s}\}$

$L_2 \rightarrow \{w \in \{0, 1\}^* \mid w \text{ contains 0 in every second and third position}\}$

$L_3 = \{w \in \{0, 1, 2\}^* \mid 0^i 1^j 2^k : i, j, k \geq 0 \text{ and } i > j + k\}$

A. Write a CFG for L_1 . (3 Marks)

Answer:

$S \rightarrow 0S1S \mid 1S0S \mid \epsilon$

OR

$S \rightarrow 0S1 \mid 1S0 \mid SS \mid \epsilon$

B. Write a CFG for L_2 (4 Marks)

Answer:

For example: 100000100, where every group has the form x00.

$S \rightarrow A00S \mid A0 \mid A \mid \epsilon$

$A \rightarrow 0 \mid 1$

D. Write a CFG for L_2 . (3 Marks)

Answer:

$S \rightarrow 0S \mid 0A$

$A \rightarrow 0A2 \mid B$

$B \rightarrow 0B1 \mid \epsilon$